Quantum Extension of ARQ: Entanglement as a Value Signal

Technical note on resource-aware quantum error valuation

Core claim. In a quantum setting, an error event should not be valued only by immediate damage. It should also be valued by whether it creates *usable entanglement* inside the trusted control boundary of the system. Some errors are pure loss. Some are accidental discoveries.

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Place and date	Bruxelles, 2026
Document type	Technical note / conceptual extension

Abstract

This note extends ARQ into a quantum regime, where an error event is treated as a state transition that may either destroy or create future computational value. The key distinction is between **destructive entanglement** with an uncontrolled environment and **constructive entanglement** with trusted ancillas that remain within the operational boundary of the entity. The value function is therefore extended with a term that measures the change in usable entanglement, allowing some apparent errors to be reclassified as experience artifacts rather than pure loss.

1. State Representation

Let the entity c be associated with a density matrix ρ over a composite Hilbert space

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E,$$

where $\mathcal{H}_S$ is the system of interest, such as a quantum memory or cognitive register, and $\mathcal{H}_E$ is the environment, including noise sources, measurement apparatus, and ancillas outside the trusted control boundary.

In quantum error correction, a pure state is fragile and a mixed state carries uncertainty. ARQ interprets an error event as a transition

$$\rho_{\text{pre}} \longrightarrow \rho_{\text{post}}.$$

The event is not judged only by immediate fidelity loss. It is judged by what kind of correlation structure exists after the transition, and whether that structure can still be used.

2. Entanglement as a Resource

The von Neumann entropy of the reduced system state measures how strongly the system is correlated with the rest of the world:

$$S(\rho_S) = -\operatorname{Tr}(\rho_S \log \rho_S), \qquad \rho_S = \operatorname{Tr}_E(\rho).$$

Low $S(\rho_S)$ indicates a state close to purity, hence high coherence and stronger potential for quantum advantage. High $S(\rho_S)$ indicates stronger entanglement with external degrees of freedom and usually signals decoherence.

Two operational cases

Destructive entanglement. The system becomes correlated with an uncontrolled environment and the resulting structure cannot be harnessed later. Coherence is lost without opening a recoverable computational path.

Constructive entanglement. The correlation involves trusted ancillas or controllable subsystems that remain inside the operational boundary of the entity. What first appears to be an error may actually open a path for teleportation, syndrome extraction, measurement-based computation, or future inference.

The distinction is operational rather than mystical: the same physical correlation can be either loss or resource depending on whether the correlated degrees of freedom remain under control.

3. Measuring Error Value via Entanglement

We extend the ARQ value function with a term that captures the change in usable entanglement:

$$V = \alpha\,\Delta S + \beta\,U(\rho_{\text{post}}) - \gamma\,C - \delta\,R + \varepsilon\,\Delta E_{\text{usable}}.$$

Here,

$$\Delta E_{\text{usable}} = E_{\text{usable}}(\rho_{\text{post}}) - E_{\text{usable}}(\rho_{\text{pre}}), \qquad \Delta S = S(\rho_S^{\text{pre}}) - S\left(\rho_S^{\text{post}}\right).$$

With this sign convention, $\Delta S > 0$ means the system became more ordered or more coherent, while $\Delta S < 0$ means local entropy increased.

Suitable entanglement monotones include:

$$\mathcal{N}(\rho) = \frac{\left\|\rho^{T_B}\right\|_1 - 1}{2}, \qquad E_N(\rho) = \log_2 \left\|\rho^{T_B}\right\|_1.$$

For multipartite systems, one may instead use tangle, robustness, or a geometric entanglement measure.

Interpretation

An event that increases E_{usable} may be valuable even when local entropy rises. An event that only increases uncontrolled entanglement with the environment is destructive. This prevents the simplistic rule that all deviations are necessarily bad.

4. Practical Computation of E_{usable}

Assume the quantum entity has a known set of trusted ancillas $\mathcal{A}_{\text{trusted}}$. Then a practical definition is

$$E_{\text{usable}}(\rho) = \max_{A \subseteq \mathcal{A}_{\text{trusted}}} \mathcal{N}(\rho_{SA}).$$

Here ρ_{SA} is the reduced state over the system together with a chosen subset of trusted ancillas. The maximisation over subsets captures the best quantum advantage that can still be extracted from the controlled part of the architecture.

In practice, full tomography may be too expensive, so one often uses approximations: concurrence on selected qubit pairs, fidelity with Bell states, witness operators tailored to the expected circuit family, or compressed-sensing methods such as shadow tomography.

5. Anti-Resonance in Quantum Systems

Anti-resonance pulses, including dynamical decoupling schemes, aim to disentangle the system from the uncontrolled environment without destroying constructive entanglement inside the trusted set.

If a pulse sequence is represented by a unitary U_{pulse}, then

$$\rho_{\text{after}} = U_{\text{pulse}}\, \rho_{\text{before}}\, U_{\text{pulse}}^{\dagger}.$$

The optimisation target is not simply to minimise entropy at any cost. The real objective is to minimise $S(\rho_S)$ while preserving or increasing E_{usable}. This becomes a constrained offline optimisation problem for a known noise model, device topology, and trusted-ancilla map.

6. Example: Bell Pair Creation as a Valuable Error

Suppose the pre-event state is a product state

$$\rho_{\text{pre}} = |0\rangle\langle 0| \otimes \sigma_{\text{ancilla}}, \qquad \rho_{\text{post}} = |\Phi^+\rangle\langle\Phi^+|,$$

where $|\Phi^+\rangle$ is a Bell state between the system qubit and a trusted ancilla.

Then the reduced system entropy increases from 0 to 1 bit, so $\Delta S < 0$ under the convention above. At the same time, E_{usable} jumps from 0 to 1 when logarithmic negativity is used. If the Bell pair is later consumed for teleportation, entanglement swapping, or superdense coding, then $U(\rho_{\text{post}})$ is high.

Hence the overall value can still become positive, and the event is promoted from mere error to an **Experience Artifact**: something the entity should remember, replay, and stabilise deliberately next time.

7. Security and Witnessing

Every quantum error event that changes E_{usable} should be recorded in a tamper-evident witness trail. The log should include at least the following:

1. pre- and post-event state estimates, obtained through tomography, shadows, or compressed sensing;
2. the pulse sequence or recovery operation that was applied, if any;
3. the computed value of E_{usable} before and after the event;
4. the valuation output V together with the chosen coefficients and device assumptions;
5. signatures or attestation artifacts from a trusted execution boundary, when the quantum hardware exposes such a feature.

This gives the theory an audit spine. A valuable error cannot remain a mystical claim; it must be replayable, challengeable, and attributable.

8. Operational Grounding

Grounding note

Real hardware is rude, and that is healthy.

On NISQ devices, state tomography scales badly, calibration drifts, pulse schedules leak into each other, and the line between trusted ancilla and environment may move with temperature, crosstalk, or firmware updates. Because of that, E_{usable} should be treated as a bounded engineering estimate rather than a divine quantity.

The physically relevant question is not “is there entanglement somewhere?” but “can this exact hardware stack reproduce, verify, and exploit it before noise, time, and cost erase the advantage?” That grounding matters for ARQ. A correlation that exists only on paper has little value. A correlation that survives measurement budget, control latency, and verification cost is another story entirely.

Closing note. The quantum extension of ARQ reframes error valuation from a purely destructive model into a resource-aware one. In this view, an error is judged not only by deviation from the intended path, but by whether it damages recoverable structure or accidentally creates it. Entanglement becomes a value signal precisely when it remains inside a controlled and witnessable boundary.

Notation at a glance

Symbol	Meaning
ρ	Density matrix of the full system-environment state.
ρ_S	Reduced state of the system of interest after tracing out the environment.
$S(\rho_S)$	Local von Neumann entropy, used here as a coherence-loss indicator.
E_{usable}	Recoverable entanglement inside the trusted control boundary.
$U(\rho_{\text{post}})$	Downstream utility of the post-event state for future computation or inference.
C and R	Operational cost and risk terms attached to the event or recovery path.
V	ARQ valuation score for promoting, rejecting, or remembering the event.

Minimal witness record

A useful implementation baseline is simple: record the pre-state estimate, post-state estimate, applied pulse or recovery schedule, computed E_{usable} before and after, and a signed hardware or controller attestation when available. Without that trail, the claim that an error was valuable is too soft for engineering use.